

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n, head, seek_time = 0;
6
7      printf("Enter the number of disk requests: ");
8      scanf("%d", &n);
9      int request_queue[n];
10
11     printf("Enter the disk request queue:
12 ");
13     for (int i = 0; i < n; i++) {
14         scanf("%d", &request_queue[i]);
15     }
16
17     printf("Enter the initial position of the disk head: ");
18     scanf("%d", &head);
19
20     // Sort the request queue to simplify SCAN algorithm
21     for (int i = 0; i < n - 1; i++) {
22         for (int j = i + 1; j < n; j++) {
23             if (request_queue[i] > request_queue[j]) {
24                 int temp = request_queue[i];
25                 request_queue[i] = request_queue[j];
26                 request_queue[j] = temp;
27             }
28         }
29     }
30
31     // SCAN (Elevator) Disk Scheduling
32     printf("
33     SCAN (Elevator) Disk Scheduling:
34 ");
35     int start = 0;
36     int end = n - 1;
37     int current_direction = 1; // 1 for moving right, -1 for moving left
38
39     while (start <= end) {
40         if (current_direction == 1) {
41             for (int i = start; i <= end; i++) {
42                 seek_time += abs(i - head);
43                 head = i;
44             }
45             current_direction = -1;
46         } else {
47             for (int i = end; i >= start; i--) {
48                 seek_time += abs(i - head);
49                 head = i;
50             }
51             current_direction = 1;
52         }
53         start = head;
54         end = head;
55     }
56
57     printf("Total Seek Time: %d\n", seek_time);
58     printf("Average Seek Time: %.2f\n", (float)seek_time / n);
59 }
```

Enter the number of disk requests: 5

Enter the disk request queue:

82

170

43

140

24

Enter the initial position of the disk head: 50

SCAN (Elevator) Disk Scheduling:

Total Seek Time: 120

Average Seek Time: 24.00

=== Code Execution Successful ===

